A LOCALLY NOETHERIAN SPACE IS LOCALLY CONNECTED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The decomposition of a Noetherian space into finitely many irreducible components, and the inheritance of Noetherianity by subsets. Invoked inside the counted proof (source lines 1591-1596). Supplies the finite decomposition of a Noetherian neighbourhood.

Source reference. Original lemma lemma-Noetherian: sources/stacks-topology/pinned-original-source.tex, source lines 1470–1470. Navigation pointer only: the counted text refers to the Stacks item lemma-Noetherian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a locally Noetherian topological space. Then X is locally connected.*

Proof. Let $x \in X$. Let E be a neighbourhood of x. We have to find a connected neighbourhood of x contained in E. By assumption there exists a neighbourhood E' of x which is Noetherian. Then $E \cap E'$ is Noetherian, see Lemma lemma-Noetherian. Let $E \cap E' = Y_1 \cup \ldots \cup Y_n$ be the decomposition into irreducible components, see Lemma lemma-Noetherian. Let $E'' = \bigcup_{x \in Y_i} Y_i$. This is a connected subset of $E \cap E'$ containing x. It contains the open $E \cap E' \setminus (\bigcup_{x \notin Y_i} Y_i)$ of $E \cap E'$ and hence it is a neighbourhood of x in X. This proves the lemma. □

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